TẠP I

Session 24

Instructor: Hồ Lê Vũ

Start recording...

OVERVIEW

- SKIN CARE
- Class Wrap-up

Slide No **4**

SKIN AGING FACTORS

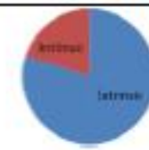
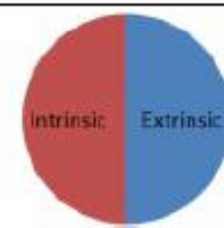
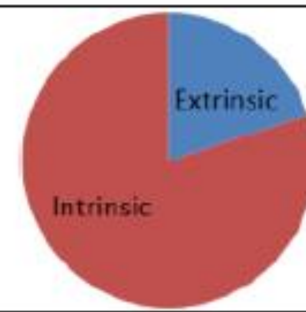
- Skin aging factors:
  - Intrinsic: genetics and cell senescence, repetitive muscle movement and loss of subcutaneous fat, bone and estrogen
  - Extrinsic: exposure to sun light, smoking and pollution

→ extrinsic aging takes place far sooner than intrinsic aging

→ *watch annotation clips*

Slide No **5**

SKINCARE ROUTINE

Age Group	Function		
	Prevention	Treatment	Anti-aging
Teen – mid 20s	FULL TIME	AS NEEDED	
Mid 20s – mid 40s	FULL TIME	AS NEEDED	
Post mid 40s	FULL TIME	AS NEEDED	

SKINCARE ROUTINE

Product Category	**Function**			
	Prevention	Treatment	Anti-aging	
			Extrinsic	Intrinsic
Cleansing	✓	✓		
Toning	?	?		
Sun protection	✓	✓	✓	
Exfoliation	✓	✓	✓	
Anti-oxidation	✓	✓	✓	
Topical probiotic/prebiotic	✓	✓	?	
Moisturizing	?	✓		
Acne treatment		✓		
Skin lightening		✓		
Tech-based treatments		✓	?	

CLEANSING

- gentle cleaning ~~deep-cleaning~~ ~~soap & bar cleansers~~
  - Double cleansing: (an oil-based cleanser + a water-based cleanser) for heavy/waterproof make-up or sunscreen
- Mechanics: mostly surfactant-based product
  - Very dry or sensitive skin: rinse-free cleansers with non-traditional ester-based surfactants → work as double-cleansers too
    - Phospholipids in micellar water products

CLEANSING

- Purpose singularity: no ingredients for ~~leave-on~~ products
- Cost effectiveness
  - electronic facial cleansing brush (with or without massaging functions)

FOOD FOR THOUGHT

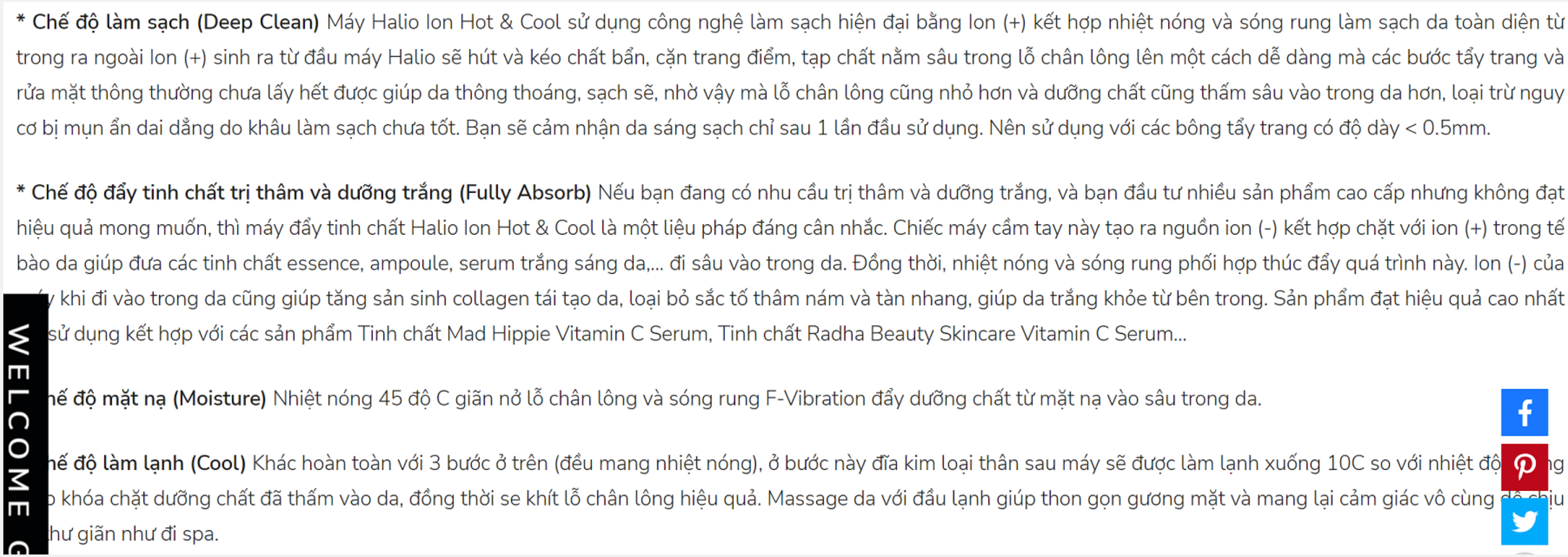
1. Công nghệ Ion Galvanic kết hợp nhiệt nóng 45 độ C: kích thích các dưỡng chất thấm sâu vào hạ bì da, tăng gấp 3.5 lần hiệu quả thẩm thấu dưỡng chất so với thoa tay thông thường

2. Sóng rung F-Vibration + xung điện EMS : cải thiện tuần hoàn máu, nâng cơ và làm săn chắc da

3. Chế độ làm lạnh: mát xa da thư giãn, khóa dưỡng và se khít lỗ chân lông

4. 4 chế độ làm đẹp, 5 cấp độ mạnh nhẹ, an toàn cho cả làn da nhạy cảm.

FOOD FOR THOUGHT


*** Chế độ làm sạch (Deep Clean)** Máy Halio Ion Hot & Cool sử dụng công nghệ làm sạch hiện đại bằng Ion (+) kết hợp nhiệt nóng và sóng rung làm sạch da toàn diện từ trong ra ngoài Ion (+) sinh ra từ đầu máy Halio sẽ hút và kéo chất bẩn, cặn trang điểm, tạp chất nằm sâu trong lỗ chân lông lên một cách dễ dàng mà các bước tẩy trang và rửa mặt thông thường chưa lấy hết được giúp da thông thoáng, sạch sẽ, nhờ vậy mà lỗ chân lông cũng nhỏ hơn và dưỡng chất cũng thấm sâu vào trong da hơn, loại trừ nguy cơ bị mụn ẩn dai dẳng do khâu làm sạch chưa tốt. Bạn sẽ cảm nhận da sáng sạch chỉ sau 1 lần đầu sử dụng. Nên sử dụng với các bông tẩy trang có độ dày < 0.5mm.

*** Chế độ đẩy tinh chất trị thâm và dưỡng trắng (Fully Absorb)** Nếu bạn đang có nhu cầu trị thâm và dưỡng trắng, và bạn đầu tư nhiều sản phẩm cao cấp nhưng không đạt hiệu quả mong muốn, thì máy đẩy tinh chất Halio Ion Hot & Cool là một liệu pháp đáng cân nhắc. Chiếc máy cầm tay này tạo ra nguồn ion (-) kết hợp chặt với ion (+) trong tế bào da giúp đưa các tinh chất essence, ampoule, serum trắng sáng da,... đi sâu vào trong da. Đồng thời, nhiệt nóng và sóng rung phối hợp thúc đẩy quá trình này. Ion (-) của ...y khi đi vào trong da cũng giúp tăng sản sinh collagen tái tạo da, loại bỏ sắc tố thâm nám và tàn nhang, giúp da trắng khỏe từ bên trong. Sản phẩm đạt hiệu quả cao nhất ... sử dụng kết hợp với các sản phẩm Tinh chất Mad Hippie Vitamin C Serum, Tinh chất Radha Beauty Skincare Vitamin C Serum...

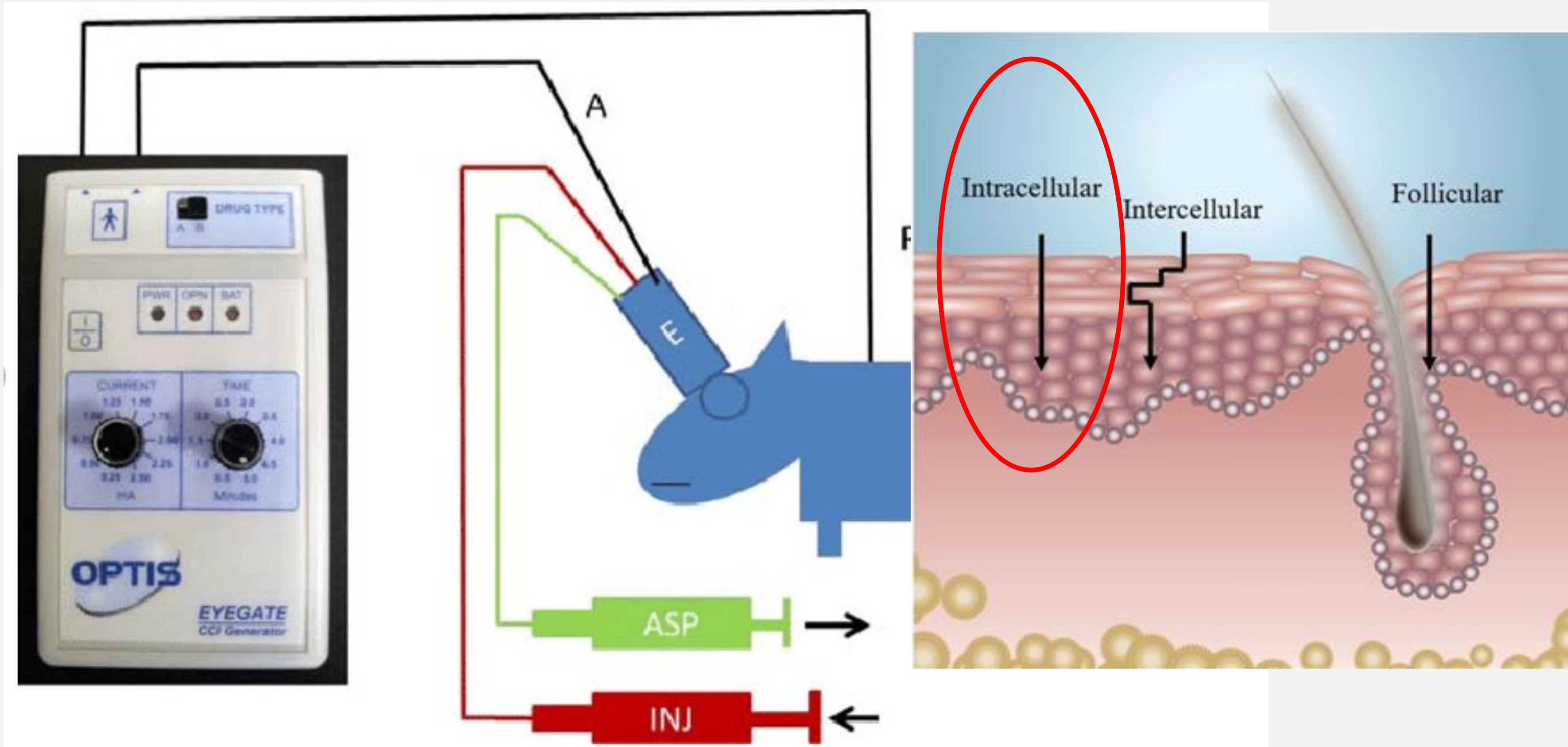
...hế độ mặt nạ (Moisture) Nhiệt nóng 45 độ C giãn nở lỗ chân lông và sóng rung F-Vibration đẩy dưỡng chất từ mặt nạ vào sâu trong da.

...hế độ làm lạnh (Cool) Khác hoàn toàn với 3 bước ở trên (đều mang nhiệt nóng), ở bước này đĩa kim loại thân sau máy sẽ được làm lạnh xuống 10C so với nhiệt độ ...ng ...o khóa chặt dưỡng chất đã thấm vào da, đồng thời se khít lỗ chân lông hiệu quả. Massage da với đầu lạnh giúp thon gọn gương mặt và mang lại cảm giác vô cùng d... ...iu ...hư giãn như đi spa.

Iontophoresis technology?

Slide No **11**

FOOD FOR THOUGHT



Iontophoresis device. A translimbal applicator adapted to the size of the rat cornea is filled with the solution to be administered and connected to the negative electrode. The positive needle electrode is connected to the neck skin as a return electrode. Both electrodes are connected to the CCI generator.

FOOD FOR THOUGHT

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WELCOME

Công nghệ Mỹ?
FDA-approved?

FOOD FOR THOUGHT

TONER?

FOOD FOR THOUGHT

Can you shrink pores?

No, you can't shrink them, open them or close them. When it comes to pore-control products you'll see that they claim to reduce the appearance of large pores (not actually make pores smaller). That may sound like a subtle distinction but it's really not. There's not much you can do to physically make your pores smaller, but you can avoid making them look larger.

What makes pores look large?

- Dirt - Pores get clogged with skin debris, makeup and dirt from the environment.
- Oil - Excessive oil can keep pores filled with a layer of oil that accentuates their appearance.
- Bacterial Growth - Contributes to blackhead growth, which makes pores look big.
- Sun Exposure - Can thicken the skin cells around the edge of pores, making them appear larger.
- Genetics - Determines your skin type and if you`re born with oily, thicker skin your pores will probably be more noticeable.

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FUN ACTIVITY

→ *Makeover tricks for guys*

SUN PROTECTION

WHAT UV LIGHT DOES TO SKIN TO CAUSE SUNBURN, WRINKLES AND CANCER *by* **Bernie Hobbs**

UV light causes everything from sunburn to skin cancer and the wrinkles and sagging that come with age. But what exactly does it do in our bodies to **wreak all that havoc**?

Ultraviolet (UV) light is just a higher energy version of the light we see (visible light). It's that higher energy that makes the UV in sunlight damaging to our cells and tissues. To do any damage, UV light has to be absorbed. This happens at the molecular level and it happens one electron at a time. **When a single electron absorbs a photon of UV light, that electron goes into a higher energy state. An excited electron like that makes a molecule behave in different ways** — sticking to things it shouldn't stick to, changing shape, and generally **messing with normal healthy cell business**.

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TEXT

Luckily we evolved in a world that's **saturated with** UV light, so our cells have built-in repair kits for the damage UV radiation can do. But **if the damage outweighs our capacity to repair**, or our repair kits themselves get damaged, it's **hello to liver spots** and — way too often — skin cancer.

There are two kinds of UV radiation in sunlight that reach us — UVA and the higher energy UVB (the ozone layer absorbs all UVC). And the trouble starts when they're absorbed by our more important molecules — DNA, RNA or proteins.

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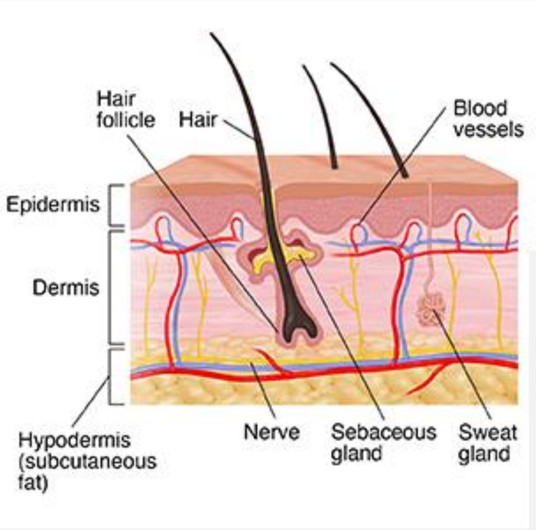
TEXT

UVB can **break proteins apart**, and **the high-energy molecules** that result **(a.k.a., radicals) are spectacular at attacking DNA**. We **need anti-oxidants to soak up these destructive radicals**.

While cancer and sunburn[1] are likely to be result of UVB, UVA is almost entirely responsible for the wrinkles and other effects of ageing on our skin as it can **penetrate more deeply into the dermis** (the second layer of skin cells) where it converts firm, youthful skin to something that looks like mine.

You can **get sunburned on a cloudy day as UVB radiation can penetrate clouds**. **UV won't burn you through a window, but will age you as glass absorbs all UVB, but not all UVA.**



TEXT

Anyone who's spent more than $10 on a beauty product knows **the key to firm skin is a protein called collagen**. It comes in long fibers that form a mesh, giving structure to our flesh. **UVA doesn't directly attack the collagen**. Instead, it **activates receptors that produce the enemy of firm skin: a particular kind of enzymes that break down collagen**. And **it doesn't take much UV at all to get this going, so even without sunburn any parts of your skin that are exposed to the sun will age.**

As if the collagen attack wasn't insult enough, **UV radiation also interferes with the production of Vitamin A receptors on our skin cells**. **Vitamin A is critical for cell growth in our skin**, but without functioning receptors for the vitamin to activate, **our skin ends up thinning**, and that's something no amount of carrots can fix.

TEXT

THE EFFECTS OF BLUE LIGHT ON SKIN *by* **Bridget March**

It seems protection from UVA and UVB rays is no longer enough, as evolving research confirms that **blue light** (also referred to as **High Energy Visible** or HEV light, generally defined as visible light from **380 to 500 nanometers**) **– emitted from the sun and our digital devices – is also bad for our skin**. We are now seeing increasing data on the potential long-term harms of visible light, and in particular blue light, on our skin. It comes as we're increasingly attached to **our digital devices**, swiftly being labeled **the silent agers of our generation**.

TEXT

Without question, sunlight is the main source of blue light to which we are exposed; digital devices **emit** only a fraction of that amount of radiation. A major difference is that **our phones are much closer to us than the sun**, and this "close-up" exposure matters. With millennials checking their phones an average of 150 times a day, and US adults clocking up more than 10 hours of screen time daily, we're getting significantly more blue light exposure than we used to. **Spending four eight-hour workdays in front of a computer exposes you to the same amount of energy as 20 minutes in the mid-day sun**.

Blue light poses potential skin harm via free radical generation, inducing oxidative stress in live skin. However blue light has the ability to **penetrate deeper into the skin compared with both UVA and UVB** light, in fact, **all the way to our dermis, where our collagen and elastin live**. Read: wrinkling and sagging as a result. Another way blue light might be ageing us is by causing pigmentation as studies indicate that **visible light may also be more active in inducing pigmentation compared to UV light**.

TEXT

Blue light also compromises our beauty sleep by way of disrupting our natural circadian rhythm. If exposed to significant amounts of blue light at night, you may find it more difficult to fall asleep, since blue light affects the level of melatonin, our sleep hormone. Not only this, but new research demonstrates that blue light exposure at night impacts the natural circadian rhythm of skin cells themselves, throwing it 'out-of-sync', **causing skin cells to continue to 'think' it is daytime, impacting their natural nighttime repair process**, which can lead to **visible signs of ageing**, and even **dark under-eye circles**.

To protect your skin from blue light damage, **cover your phones and computers with a blue light shield**. Smart phones have a setting that **disables blue light in favor of yellow light (a.k.a., 'night mode'**), which makes it easier on your eyes and on your skin. Use it all the time. When it comes to protection from skincare products, **topical antioxidants are "an absolute must".**

FINAL POINTS

- Sun exposure causes > 80% of extrinsic aging

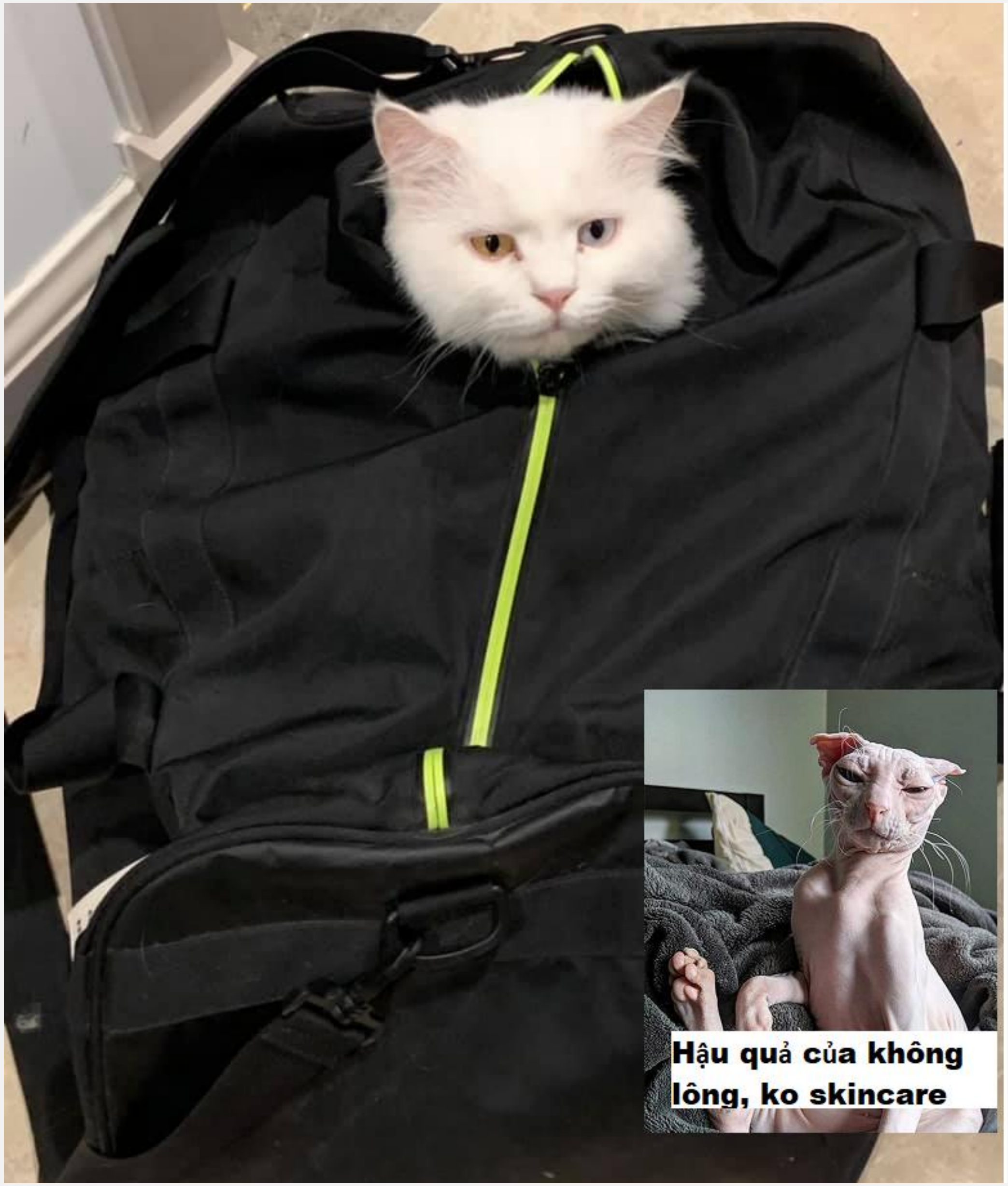
SUN PROTECTION

- BROAD-SPECTRUM protection against UVB & UVA
- UVB protection: >= SPF 30 (sun protection factor)
  - If it took *n* minutes for redness to appear in unprotected skin → 30 times that for the same symptoms to appear in the skin protected by an SPF 30 sunscreen (in an ideal condition)
  - NO 100% UVB blocking: SPF 30 blocks 97%; SPF 50 - 98%; SPF 100 - 99% → *should the the more the merrier principle apply?*

SUN PROTECTION

- UVA protection:
  - currently no globally standardized protection factor for UVA
    - PPD (Persistent Pigment Darkening, Asia & Europe): a PPD of 10 = 10 times longer for your protected skin to tan
    - PA: PA+ (PPD of 2 to < 4), PA++ (4 to 8), PA+++ (8 to 16) & PA++++ (>= 16)
- Use other types of **physical protection**: hat, clothes, face mask and shades

FOOD FOR THOUGHT



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FDA-REGULATED SUNSCREEN

fda.gov/drugs/understanding-over-counter-medicines/sunscreen...

Sunscreen ingredients

Below is a list of acceptable active ingredients in products that are labeled as sunscreen:

Aminobenzoic acid	Octisalate
Avobenzone	Oxybenzone
Cinoxate	Padimate O
Dioxybenzone	Ensulizole
Homosalate	Sulisobenzone
Meradimate	Titanium dioxide
Octocrylene	Trolamine salicylate
Octinoxate	Zinc oxide

Two proven-effective UVA blockers
chemical Avobenzone *vs.* physical Zinc Oxide

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NANO TECHNOLOGIES

- insoluble nanopigments as UV-filters to reflect UV light

SUN PROTECTION

- EWG's 2024 Guide to Sunscreens rates safety, efficacy of more than 1700 SPF products
  - about 75% have inferior sun protection or worrisome ingredients
  - list of sunscreens measure up to EWG's rigorous standards: https://www.ewg.org/sunscreen/
- The guide's best-scoring sunscreens contain zinc oxide, titanium dioxide or both

EWG'S TIPS

- Avoid products with oxybenzone, which is absorbed through the skin in large amounts and can affect hormone levels.
- Stay away from vitamin A in sunscreens. Government studies link the use of retinyl palmitate, a form of vitamin A, to the formation of skin tumors and lesions when it's applied to sun-exposed skin.
- Steer clear of sunscreens with SPF values above 50+, which may not give increased UVA protection and can fool people into thinking they're safe from sun damage.
- Avoid sprays. These popular products make it difficult to apply an adequate and even coating on skin, especially in windy conditions. They also pose inhalation concerns.
- Avoid intense sun exposure during peak hours for sun exposure, between 10 a.m. and 4 p.m.

FOOD FOR THOUGHT

Myths and Realities About Fun in the Sun

MYTH:	REALITY:
A suntan is healthy. Establishing a base suntan protects you from sun damage.	A tan results from the body defending itself against further damage from UV radiation. Any change in your skin's natural color is a sign of damage to the skin.
You cannot get a sunburn on a cloudy day.	Sunburn is possible on a cloudy day. Up to 80% of solar UV radiation can penetrate light cloud cover.
UV radiation during winter is not a concern.	UV radiation is generally lower in winter, but snow reflection can double overall exposure — especially at high altitudes — leading to sunburn and snowblindness.
Sunscreen protects you so that you can sunbathe much longer.	Sunscreen should not be used to increase sun exposure time but to increase protection during unavoidable exposure.
If you take breaks while sunbathing, you won't get sunburn.	UV radiation exposure is cumulative during the day.
If you don't feel the sun's hot rays, you won't get sunburn.	Sunburn is caused by UV rays that cannot be felt. The heating effect is caused by the sun's infrared radiation and not by UV.
Skin cancer only occurs on parts of the body that are exposed to the sun all the time.	Melanoma occurs most commonly on the back (for men) and legs (for women), which are sites with only intermittent exposure.
Skin cancer only happens to people who are very fair-skinned.	Skin cancer commonly occurs in people who tan before they burn.

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SUN PROTECTION

Check UV index

weather.com/weather/hourbyhour/l/25daed82151984c9148

Hourly Weather - Cát Linh, Hanoi, Vietnam

As of 11:29 am

Tuesday, May 10

12 pm **31°** Partly Cloudy

Feels Like **38°**

Wind **SSW 10 km/h**

UV Index **Extreme**

Cloud Cover **54%**

1 pm **32°** Mostly Clou

2 pm **32°** Mostly Clou

3 pm **32°** Cloudy

4 pm **31°** Cloudy

Feels Like **38°**

Wind **ESE 9 km/h**

UV Index **1 of 10**

Cloud Cover **90%**

https://en.tutiempo.net/ultraviolet-index/

World UV Index

Ultraviolet radiation hour by hour

Ultraviolet Index in ...

The UV index or UV radiation comes directly from the sun and we are all exposed to it. The UV region spans the wavelength range from 100 to 400 nm and is divided into the following three bands:

- UVA (315-400 nm)
- UVB (280-315 nm)
- UVC (100-280 nm)

SUN PROTECTION & MAKEUP

https://www.stylist.co.uk/beauty/spf-foundation-moisturiser-best-sunscreen/261637

SPF has long had a bad reputation as far as formulations go, though. It can be too thick, chalky, greasy and leave skin looking ashy or with a white cast (although, we must say products have come on a long way since then). So, it's no wonder that so many people admit that instead of having a separate SPF, they just rely on what's already built into their daily moisturiser, foundation or BB cream to protect their skin. Sadly – and we hate to be the ones to break it to you – that isn't going to give you adequate protection against the sun's UVA and UVB rays, mostly because it's rare that enough product will be applied.

Most experts agree that in order to get the required amount of protection from sunscreen, around 1-1.5tsps needs to be applied, which is much more than the amount of foundation anybody really uses on a daily basis. "Instead, make-up with SPF should be thought of as your second line of defence against UV rays once you've already applied sunscreen," adds Dr Kluk. "Because two forms of sun protection are much better than one."

SUN PROTECTION & VITAMIN D

SUN PROTECTION AND VITAMIN D *by* **Anne Marie Mcneill**, MD, PHD *&* **Erin Wesner**

You need sun protection as much as you need vitamin D. You can have both, without skin damage or nutritional deficiency*. A dermatologist tells you how.*

Vitamin D helps keep your bones strong by regulating calcium levels. Maintaining adequate amounts of the vitamin is essential for your bone health. People deficient in the vitamin can suffer symptoms including muscle aches, muscle weakness and bone pain. In severe cases, by reducing calcium absorption, deficiency can lead to slower growth, bone softening and weakened bone structure, increasing the risk of skeletal deformities, osteoporosis and fractures.

If you're having blood drawn for your annual checkup, ask your doctor to test your vitamin D level. On your lab report, here's what your number means.

Below 30: Deficient. Talk to your doctor about supplements.

30 to 50: Generally inadequate for bone and overall health.

50 to 125: Adequate (but **more is not necessarily better**).

125 and above: Too high (may have adverse effects).

When your skin is exposed to sunlight, it manufactures vitamin D. The sun's ultraviolet B (UVB) rays interact with a protein called 7-DHC in the skin, converting it into vitamin D3, the active form of vitamin D.

FUN ACTIVITY

- *A tanning incident…*

EXFOLIATION

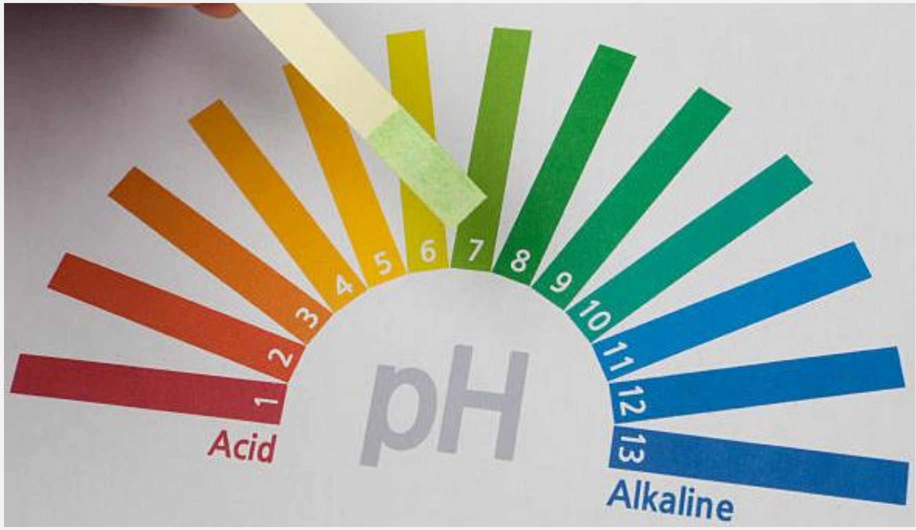
- Excess of unshed surface skin cells → a flaky & uneven skin tone with clogged and enlarged pores
- Removing layers of dead skin cells to
  - improve skin texture
  - unclog pores and prevent acne
  - allow other ingredients to be better absorbed

EXFOLIATION

- AHAs (e.g., glycolic & lactic acids) vs. BHA (salicylic acid): water soluble *vs.* lipid soluble
  - AHAs (skin discolorations & collagen production) → more suitable for sun-damaged, thickened, or dry skin.
  - BHA (pore penetration & bacteria killing) → the best choice for blackheads and blemishes

EXFOLIATION

- Concentration & pH
  - AHAs: 5% to 8%, a pH of 3 to 4 (and not greater than 4.5)
  - BHA: 1% to 2%, a pH of 3 (and not greater than 4)



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FOOD FOR THOUGHT

- Natural forms of AHA or BHA cannot exfoliate skin
  - Pure water (neutral, neither acidic nor alkaline): pH of 7
  - Sugarcane juice (4.6 and above)
  - Milk is slightly acidic (6.3 and above)
  - Yogurt is more acidic than milk (about 4.4, due to the conversion of milk sugar or lactose into lactic acid)
  - *Fun fact*: skin is naturally acidic (between 4.7 & 5)

EXFOLIATION

- Sensitivity risk (slight stinging and initial redness are expected)
  - AHAs could be irritating & BHA drying to sensitive skin
- Alternatives for sensitive skin:
  - AHA
    - Mandelic acid
    - PHA
  - BHA
    - LHA

EXFOLIATION ALTERNATIVES

- stronger exfoliants, e.g., ~~phenol~~
- mechanical exfoliants (scrubs, brushes, washcloths) for less sensitive other body parts.

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EXFOLIATION ALTERNATIVES

- peeling gel & enzyme powder (cleanser-exfoliant hybrid)
  → not very effective as double-cleansers

https://lionesse.com/facial-peeling-gels-how-do-they-work-are-they-an-effective-way-to-exfoliate/

There are two main **ways to exfoliate**; physically and chemically. Facial peeling products fall in the former category. They're a physical exfoliant, yet they're far gentler than the average face scrub.

They look and feel just like a silky gel, making them quite satisfying to apply, especially when you can literally see them go to work.

How?

Well, a facial peeling gel contains special ingredients that interact with the dead skin cells and excess sebum on your skin. As you massage the product into your skin, it'll start to form tiny clumps. As you continue to gently rub, these clumps will pick up even more debris, taking all of that away with them once you wash your face.

https://inly.ua/en/delikatnoe-glubokoe-ochishenie-kak-rabotaet-enzimnaya-pudra-en

Enzyme powder is suitable for cleansing the skin after the make-up removal procedure. Remove makeup in the usual way, using milk or micellar water.

Apply a small amount of powder (no more than half a heaping teaspoon) to the palm of your hand, mix with a small amount of water at a comfortable temperature and rub in your palms until a gentle emulsion is achieved.

Apply the resulting soft foam to a moisturized face, neck and décolleté.

Gently cleanse the skin in circular motions - as during the procedure of washing with the usual means.

Rinse off the remaining product with water at a comfortable temperature.

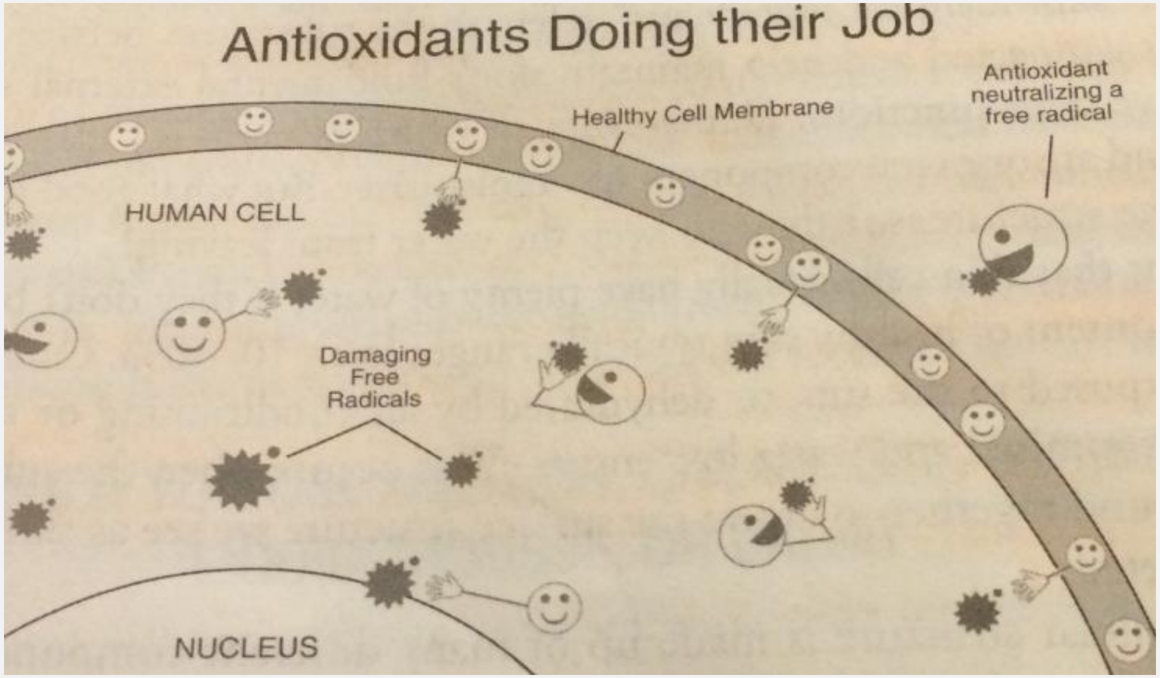
combining facial cleansing, exfoliation & massaging

OXIDATION

- Free radicals cause damage to the cell's DNA → oxidative stress → diseases and accelerated ageing
- Oxidative stress in the skin
  - Saggy skin (due to damaged elastin fibers)
  - Wrinkles (due to the breakdown of collagen)
  - Brown spots or pigmentation (due to increased melanin production)

ANTIOXIDATION

- Free-radicals neutralized by antioxidants



ANTIOXIDANTS

- Vitamin C: improve firmness, smoothness & skin tone

WHAT IS THE BEST VITAMIN C FOR YOUR SKIN

Source: https://photozyme.com/blogs/news/tetrahexyldecyl-ascorbate-the-best-vitamin-c-for-your-skin

Vitamin C products have gained superhero status amongst skincare enthusiast, and for good reason. Vitamin C is a powerhouse ingredient with numerous skin benefits that range from visibly fading dark spots to minimizing the appearance of wrinkles. So, what's not to love? Well, there's a caveat every savvy, skin-conscious consumer should be aware of before investing hard earned dollars into just any Vitamin C product. That's where Tetrahexyldecyl ascorbate comes in to play.

Stabilized Vitamin C (THDA)	VS	Common Vitamin C L-ascorbic Acid
Oil Soluble Penetrates the dermis and epidermis		**Water Soluble** Poor Skin Penetration
Stabilized Increased shelf life and prolonged effectiveness		**Unstable** Light and air rapidly decrease shelf-life and effectiveness
Non-Irritating Optimal pH for all skin including sensitive		**Irritating** Requires low pH, more likely to irritate
Optimal pH Perfect pH level for DNA repair enzymes to thrive		**Suboptimal pH** Too acidic for optimal DNA repair activity

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ANTIOXIDANTS

- Vitamin E: counteract harms from sun exposure

EVERYTHING VITAMIN E CAN DO FOR YOUR SKIN *by* **Mara Santilli**

***The antioxidant helps minimize sun damage and pumps up moisture*.**

Topical vitamins get plenty of praise from skin care enthusiasts, but there's an important one that's often left out of the spotlight: vitamin E. You may have heard that vitamin A (a.k.a. retinol) is the gold standard for smoothing fine lines and increasing elasticity, that vitamin B3 (in the form of niacinamide) can reduce inflammation and improve acne, and that vitamin C is a go-to for targeting hyperpigmentation. But the (alphabetical) list goes on to vitamin E, which deserves a little more attention.

There are quite a few benefits of vitamin E that your skin might appreciate, and the good news is that most people get enough of it from their diet. However, a little topical vitamin E (say, in the form of a silky serum or a rich moisturizer) can also give your complexion a boost.

What is vitamin E and how does it do its thing?

Vitamin C is well known as one of the best antioxidants for skin, but vitamin E belongs in that group too. There are a few different types of vitamin E, but **the form that's most useful to your body is alpha-tocopherol**, which you can naturally find in a bunch of different foods, including olive oil, peanuts, almonds, spinach, avocados, pumpkin, red bell pepper, asparagus, and mangoes.

When experts say that vitamin E acts as an antioxidant in the body, that means it fights cell-damaging particles (called free radicals) that are known to contribute to a variety of health issues, including autoimmune diseases, asthma, eczema, heart disease, and cancer. Those apply to the skin too, and they can **specifically help protect cells from UV light damage** (like that sunburn you accidentally got when you took a nap on the beach) and skin cancer.

When it comes to UV protection, **the role of vitamin E is both preventative and reactive**. Its antioxidant

ANTIOXIDANTS

- Vitamin B3 (niacinamide)
  - well tolerated by all skin types
  - providing a “tightened-up" look of enlarged pores by reducing oil production while encouraging the production of ceramides to improve skin elasticity)
- EGCG, Resveratrol & Curcumin
  - anti-inflammatory and calming effect
- Ferulic acid
  - enhancing stability and efficacy of other antioxidants
- Lycopene
  - promoting collagen production

ANTIOXIDANTS

A combination of different antioxidants for best effects

E.g., combination of Vit C, Vit E & ferulic acid

SOURCE OF FREE RADICALS

External

- sun exposure
- cigarette smoke
- pollutants
- deep-frying
- pesticides

Internal

- [**unavoidable**] metabolic processes
- immune system responses
- alcohol consumption
- exhaustive exercise

topical application of antioxidant serums

antioxidant rich food sources, esp. fruits and vegetables

A NOTE OF CAUTION

- Dietary supplements, esp. multi-vitamin doses?
- Oral consumption of vitamins
  - Water-soluble vitamins, e.g., C & Bs are flushed out of your body with water
  - Fat-soluble vitamins, e.g., A, D, E, & K, build up in your body for long periods of time
    - Vitamin A overdose can be bad for your bones
    - Vitamin E overdose may increase the risk of heart problems

→ *discuss an extra reading*

TEXT

ANTIOXIDATION *vs.* OXIDATION (*excerpts from* **Do You Believe in Magic The Sense and Nonsense of Alternative Medicine** *by* **Paul A. Offit, M.D.**)

Many people believe that vitamins and supplements had one property that made them **cure-alls**, a property that rivals words like *natural* and *organic* for sales impact: **antioxidant**.

Antioxidation vs. oxidation has been billed as a contest between *good* and *evil*. The battle takes place in cellular organelles called mitochondria, where **the body converts food to energy, a process that requires oxygen and so is called oxidation. One consequence of oxidation is the generation of electron scavengers called free radicals** (*evil*). **Free radicals can damage DNA, cell membranes, and the lining of arteries; not surprisingly, they've been linked to aging, cancer, and heart disease. To neutralize free radicals, the body makes its own antioxidants** (*good*).

TEXT

Antioxidants can also be found in fruits and vegetables—specifically, **selenium**, **beta-carotene**, and **vitamins A**, **C**, and **E**. Studies have shown that **people who eat more fruits and vegetables have a lower incidence of cancer and heart disease and live longer.**

The logic is obvious: if fruits and vegetables contain antioxidants—and people who eat lots of fruits and vegetables are healthier—then **people who take supplemental antioxidants SHOULD also be healthier. IN FACT, THEY'RE LESS HEALTHY.**

In 1994, the National Cancer Institute, in collaboration with Finland's National Public Health Institute, studied 29,000 Finnish men, all long-term smokers more than fifty years old. This group was chosen because they were at high risk for cancer and heart disease. Subjects were given vitamin E, betacarotene, both, or neither. The results were clear: **those taking vitamins and supplements were more likely to die from lung cancer or heart disease than those who didn't take them**—the opposite of what researchers had anticipated.

TEXT

In 1996, investigators from the Fred Hutchinson Cancer Research Center, in Seattle, studied 18,000 people who, because they had been exposed to asbestos, were at increased risk of lung cancer. Again, subjects received vitamin A, betacarotene, both, or neither. Investigators ended the study abruptly when they realized that those who took vitamins and supplements were dying from cancer and heart disease at rates 28 and 17 percent higher, respectively, than those who didn't.

In 2004, researchers from the University of Copenhagen reviewed fourteen randomized trials involving more than 170,000 people who took vitamins A, C, E, and beta-carotene to see whether antioxidants could prevent intestinal cancers. Again, antioxidants didn't **live up to** the hype. The authors concluded, "**We could not find evidence that antioxidant supplements can prevent gastrointestinal cancers; on the contrary, they seem to increase overall mortality**." When these same researchers evaluated the seven best studies, they found that death rates were 6 percent higher in those taking vitamins.

TEXT

In 2005, researchers from Johns Hopkins School of Medicine evaluated nineteen studies involving more than 136,000 people and found an increased risk of death associated with supplemental vitamin E. Dr. Benjamin Caballero, director of the Center for Human Nutrition at the Johns Hopkins Bloomberg School of Public Health, said, “This reaffirms what others have said. **The evidence for supplementing with any vitamin, particularly vitamin E, is just not there**. This idea that people have that [vitamins] will not hurt them may not be that simple.” That same year, a study published in the Journal of the American Medical Association evaluated more than 9,000 people who took **high-dose** vitamin E to prevent cancer; **those who took vitamin E were more likely to develop heart failure than those who didn’t**.

In 2007, researchers from the National Cancer Institute examined 11,000 men who did or didn’t take multivitamins. Those who took multivitamins were twice as likely to die from advanced prostate cancer.

In 2008, a review of all existing studies involving more than 230,000 people who did or did not receive supplemental antioxidants found that vitamins increased the risk of cancer and heart disease.

TEXT

On October 10, 2011, researchers from the University of Minnesota evaluated 39,000 older women and found that those who took supplemental multivitamins, magnesium, zinc, copper, and iron died at rates higher than those who didn't. They concluded, "Based on existing evidence, we see **little justification for the general and widespread use of dietary supplements**."

Two days later, on October 12, researchers from the Cleveland Clinic published the results of a study of 36,000 men who took vitamin E, selenium, both, or neither. They found that those receiving vitamin E had a 17 percent greater risk of prostate cancer.

How could this be? Given that free radicals clearly damage cells—and given that people who eat diets rich in substances that neutralize free radicals are healthier—why did studies of supplemental antioxidants show they were harmful?

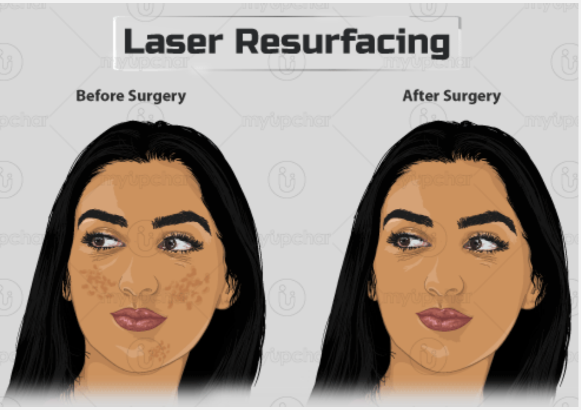
TEXT

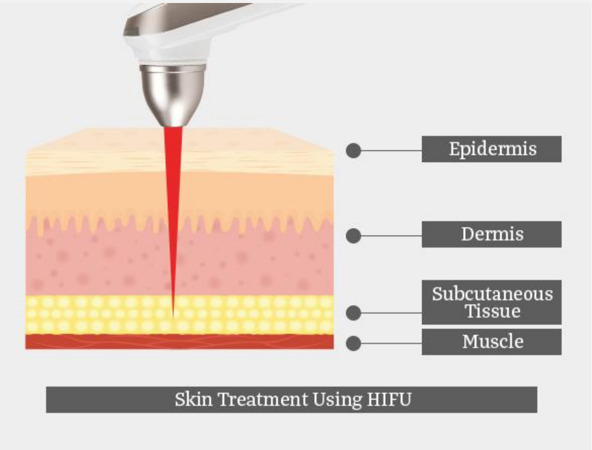
The most likely explanation is that **free radicals aren't as evil as advertised**. Although it's clear that free radicals can damage DNA and disrupt cell membranes, that's not always a bad thing. **People need free radicals to kill bacteria and eliminate new cancer cells**. But when **people take large doses of antioxidants, the balance between free radical production and destruction might tip too much in one direction, causing an unnatural state in which the immune system is less able to kill harmful invaders**. Researchers have called this "*the antioxidant paradox*."

Whatever the reason, the data are clear: **high doses of vitamins and supplements increase the risk of heart disease and cancer**; for this reason, **not a single national or international organization responsible for the public's health recommends them**.

FREE RADICAL *vs.* ANTIOXIDANT BALANCE

FOOD FOR THOUGHT





The beauty WITHIN

FUN ACTIVITY

- *Lifestyle choices vs. Life expectancy*

TEXT

Studies haven't hurt sales though. In 2010, the vitamin industry grossed $28 billion, up 4.4 percent from the year before. Despite a wealth of scientific evidence, most Americans don't know that megavitamins are unsafe. So why don't more people know about this? And why hasn't the FDA sounded an alarm? The answer is predictable: money and politics.

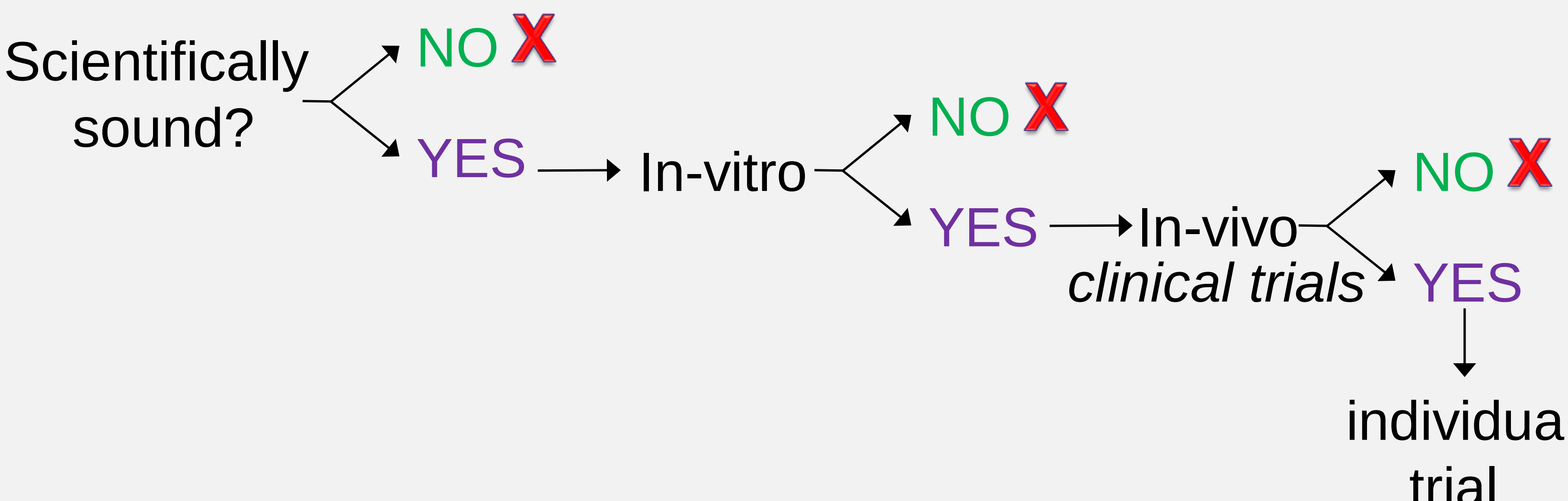
Steven Nissen, chairman of cardiology at the Cleveland Clinic, said, "**The concept of multivitamins was sold to Americans by an eager nutraceutical[1] industry to generate profits. There was never any scientific data supporting their usage.**"

Q&A

Thanh Hiền Nguyễn

Thầy ơi, thầy có thể giải thích vấn đề về uống collagen ko ạ? Em và chị em đã uống sản phẩm dạng bột. Hiệu quả là da mịn màng và sáng hơn ạ. Tuy nhưng khi ngừng uống khoảng 2 tháng thì da ko mịn nữa. Vậy cho em hỏi là collagen vẫn tác động trực tiếp đến da đúng ko ạ?

the risks of committing correlation and/or post-hoc fallacies in an uncontrolled environment with many confounding variables

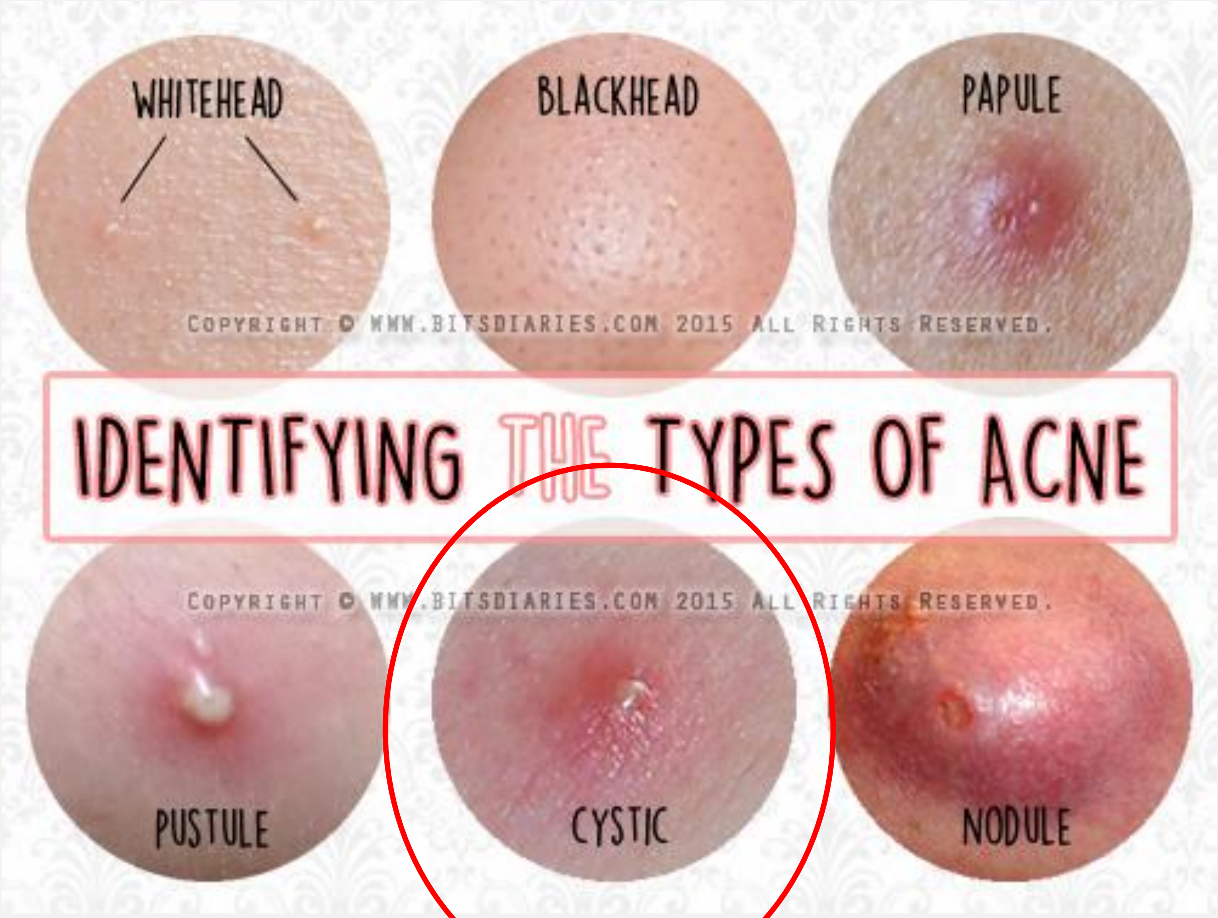


ACNE VULGARIS

- Follicles connect oil glands to the pores
- Sebum carries dead skin cells through follicles to the skin surface
- Follicles get blocked → oil builds up under the skin (non-inflammatory)
  - – in a closed pore: dead skin cells mixing with trapped oil → whitehead
  - – in an open pore: oxidation → blackhead
- A plug gets infected with bacteria (P. acnes) → swelling results (inflammatory)

ACNE TYPE

[seriousness & **depth]** papule **<** pustule/pimple **<** nodule **<** cyst



ACNE TREATMENT STEPS

1. Eliminate skin-cell build-up by effective exfoliation
2. Avoid conditions that worsen oil problems
3. Kill the P-acnes bacteria
4. Cope with oil over-production

OIL PROBLEM WORSENING

- Hot and humid climates
- Oil-based makeup
- Over-cleaning
  - Skin's perfect condition: dry and tight underneath but oily on top → over-cleaning triggers oil production

→ use oil absorbing sheet ~~alcohol-based products~~ as an alternative

P-ACNES ELIMINATION

- Topical benzoyl peroxide and/or topical antibiotics, e.g., erythromycin or clindamycin.
  - For more severe cases, oral antibiotics, e.g., tetracycline or erythromycin

P-ACNES ELIMINATION ALTERNATIVES

- As P-acnes live naturally on skin → keep skin's microbiome in balance with probiotic/prebiotic products
- Corticosteroid injection into large cysts to reduce swelling and risk of scar
- Incision and drainage of large cysts (only performed by a doctor) → risk of serious scarring and deeper infection
- High-tech treatment (e.g., light therapy, argon gas electrode)

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FOOD FOR THOUGHT

Stacked Skincare High Frequency Acne Device

$130

This high-frequency facial wand from Stacked Skincare sends an argon gas-filled electrode into the skin, which creates oxygen when it hits the skin. Bacteria cannot survive this oxygen generation, so acne-causing P. acnes are killed off. In addition to preventing pimples, it helps with plumping the skin, says Worden. It also helps reduce inflammation and redness.

MAIN CAUSES OF OIL OVER-PRODUCTION

- Genetics
- Anxiety and stress (Cortisol or stress hormone)
- Hormonal shifts (e.g., menopause, childbirth)
- Hormonal fluctuations (menstrual and/or cyclical)
- Allergic reactions to certain drugs or chemicals

OIL OVER-PRODUCTION TREATMENT

- Oral contraceptives (birth control pills)
- As the last resort: Isotretinoin (very effective for severe acne but teratogenic with many side effects)

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COVID-19 VACCINATION & ACNE

- Vaccination → the immune system in overdrive → oxidative stress → accelerate the acne process
- Getting COVID-19 vaccine and cortisone shots (injection of a synthetic hormone) at the same time?
  - the later may suppress your immune system → a risk of getting inadequate immune response → could reduce vaccine efficacy

My thoughts?

PROBIOTICS *vs.* PREBIOTICS

- Our microbiome (a.k.a., ecosystem) = combination of microorganisms (both good and bad) living within and on the surface of our bodies → should be kept in balance
  - The state of skin surface's own microbiome affect skin's appearance
- Probiotics: microbes living on skin & playing a crucial role in stabilizing the microbiome.
- Prebiotics: substances feeding and encouraging the growth and healthy balance of probiotics.
- Probiotics/Prebiotics: oral *vs.* topical application

PROBIOTICS

- e.g., Lactobacillus, Bifidabacterium & Vitreoscilla
  - found in yogurt and fermented veggies like kimchi
  - Kefir: a Russian yogurt-like drink with a diversity of bacteria (six or seven different strains of probiotics)
- offset negative factors that lead to redness, dryness & acne
- protect skin surface from environmental attack
- contribute to creation of helpful ingredients, e.g., hyaluronic acid, peptides, ceramides, and vitamins

PREBIOTICS

- plant sugars (e.g., xylitol) and fructooligosaccharides (FOS)
  - found in garlic, onions, oats, barley, asparagus, and bananas
- serve as an energy source for fragile & unstable probiotic ingredients in skincare products

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PROBIOTICS *vs.* PREBIOTICS

→ A mix of probiotics, prebiotics, and lysate ingredients (a derivative of probiotics) should be more effective (the rainforest metaphor)

- As with other organic skincare products, formulation is as important as ingredient list.

FOOD FOR THOUGHT

- Should we apply yogurt topically?
  - anti-inflammatory & anti-acne-bacteria effects of probiotics in topical use
  - psychological effects?
  - *Notes:*
    - low-sugar versions
    - acne-prone patients with oily complexions: low-fat or fat-free yogurt

FUN ACTIVITY

- *Botox injection...*

SKIN LIGHTENING

DANGERS OF SKIN WHITENING PRODUCTS *by* **Angela Clare**

Skin whitening creams contain ingredients that act to slow the production of melanin in the skin's outer layer.

Melanin is a brown pigment produced in the melanocytes in the skin. Melanin is the body's natural protection against the harmful effects of UV rays, however, an over production of melanin (hyperpigmentation) can cause uneven skin tones and other skin disorders.

Skin bleaching agents act to reduce the action of the enzyme, tyrosinase, which controls the rate of melanin production in the skin.

Pigment-reducing chemicals traditionally used in skin whitening products, namely **hydroquinine**, **mercury** and **steroids**, can **make your skin more susceptible to the sun's UV rays**. This can lead to serious sunburn and an increased risk of certain types of skin cancer.

TEXT

HYDROQUININE

Hydroquinone [C6H4(OH)2] is a **toxic chemical that is used in black and white film processing**, manufacturing rubber and is found in some hair dyes. Hydroquinine is used in skin lightening creams and lotions because it is an effective bleaching agent, slowing the production of the tyrosinase enzyme and reducing the amount of melanin formed.

While we do not know the full extent of the health risks hydroquinone poses, it is considered to be cytotoxic (toxic to cells), mutagenic and carcinogenic (cancer causing). Hydroquinine is thought to increase the risk of complications such as thyroid disorders, liver disease and adrenal dysfunction. **It has been banned for sale as a skin lightener in Europe, Japan, and Australia** and many groups are calling for the FDA to ban this chemical in the US. A ban was proposed by the FDA back in 2006 but currently **skin-lightening products that contain 2 % hydroquinone can be sold over-the-counter and products that contain up to 4 % hydroquinone can be obtained by prescription from a physician**.

Common side effects reported from the use of hydroquinine creams include skin rashes, burning skin irritation, excessive redness and a dryness or cracking of the skin. **If used for extended periods of time, hydroquinone can sometimes induce a condition known as "ochronosis." People with ochronosis can show a blue-black darkening in certain areas of the skin.**

TEXT

MERCURY

In August 2013 the Food and Drug Administration (FDA) issued a warning to consumers advising them not to use skin creams, beauty soaps or lotions that might contain mercury due to **the dangers of mercury poisoning**. **The FDA first banned the use of mercury in skin-bleaching and lightening products** back in 1990 but the regulatory office has since found that mercury was being used as an ingredient in some products that were manufactured abroad and sold illegally in the United States.

Mercury is a toxic chemical that is readily absorbed into the body but it is not easily removed. When mercury is used for skin whitening the initial side effects can include skin rashes, skin discoloration and scarring. Long term exposure to mercury can have more serious health consequences. The World Health Organisation advises that **using mercury on a long term basis can damage the kidneys and the nervous system**. It can also cause depression or psychosis and **interfere with the development of the brain in unborn children** and very young children.

The FDA recommends that you read the label of any skin lightening creams you use. **Products containing mercury will have the words *mercury*, *mercurous chloride*, *mercuric*, *mercurio*, or *calomel*.** If there is no label or list of ingredients do not use that product.

TEXT

STEROIDS

Unfortunately, many skin-lightening creams contain illegal compounds which can include high-dose steroids. Although **steroids can be useful in treating inflammation of the skin caused by diseases such as eczema, psoriasis or dermatitis**, and their prescription must take place under the supervision of a skin specialist and the usage is generally minimized to a few short weeks, **they were never intended for skin lightening use**.

While topical corticosteroids can appear to lighten the skin very quickly, this is because they act as "**vaso-constrictors**". This means **the blood vessels in the area treated will become constricted and the flow of blood will be slowed, giving the skin a whiter appearance**.

Steroids can also **slow the process of cell regeneration** so less melanocytes are formed, leading to a decline in melanin production. The negative side of slowing the skin's natural cell renewal is that the **epidermis can becomes thinner and many people complain of the appearance of green veins in the skin**.

Unmonitored use of high-dose steroids can lead to many problems. The thinning effect on the skin can increase the risk of physical damage to the skin. The skin can become more susceptible to chemical and environmental factors and there will be an increased risk of sun damage and additional pigmentation problems.

The high doses of steroids found in the illegal skin whitening creams can also interfere with the body's hormone levels and, in extreme cases, can result in disorders such as Cushings syndrome[1].

TEXT

ALTERNATIVES

Since many chemical-based skin lighteners have now been found to have serious health concerns, researchers and skin care companies have been prompted to look for safer alternatives for their consumers.

Extensive testing of many plant species has lead to the discovery of a number of all natural lightening ingredients that act to suppress melanin production, in the same way as their chemical-based counterparts, but they are non-toxic and do not carry the risk of serious side effects.

While natural skin lightening products still vary they will generally include ingredients such as **arbutin, emblica**[2]**, liquorice, mulberry extract, kojic acid**[3] **or Vitamin C**.

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SKIN LIGHTENING

- skin lighteners *vs.* skin brighteners
- drugs (hydroquinone, azelaic acid & kojic acid) *vs.* cosmetics (vitamin C, arbutin, niacinamide, etc.)

→ spot treatment (brown spots and freckles) *vs.* overall improvement (even skin tone)

→ combined usage

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DARK CIRCLE ISSUE

WHAT CAUSES DARK CIRCLES UNDER YOUR EYES? *by* **Kiara Anthony**

Dark circles under your eyes, also known as periorbital hyperpigmentation (POH), may appear as shades of brown, blue, black, or purple.

Fatigue is generally believed to be the most common cause for having dark circles under the eyes. This may be controversial, and there are actually a number of reasons.

Age

Aging may be one of the most common reasons for having dark circles under your eyes.

As you get older, **your skin tends to sag and become thinner**. You might experience a decrease in the fat and collagen that helps maintain your skin's elasticity. As this occurs, **the dark blood vessels beneath your skin become more visible, causing the area below your eyes to darken**.

Fatigue

Oversleeping, or a lack of sleep, may cause your skin to become more dull and pale. As a result, the blood vessels and dark tissues beneath your skin might become more visible.

Sleep deprivation may also lead to a **fluid buildup beneath your eyes, causing them to appear puffy**. Dark circles may then actually be shadows cast by puffy eyelids.

Eyestrain

Staring at a television or computer screen for long periods of time may strain your eyes. This strain could **enlarge the blood vessels around your eyes**, potentially causing dark circles.

Dehydration

Dehydration might attribute to POH developing. When your body isn't well hydrated, the skin beneath your eyes begins to look dull and your eyes look sunken. This is due to how close your eyes are to the underlying bone.

Sun overexposure

Prolonged sun exposure or injury may cause dark circles to form under your eyes. This is known as post-inflammatory pigmentation.

FUN ACTIVITY

- *A whitening incident...*

MOISTURIZING BASIC

- Humectant
- Occlusive
- Intercellular substance

MOISTURIZING

- Humectants (glycerin & lactic acid, etc.) attract water to skin
  - Hyaluronic acid
    - *a.k.a.*, nature's moisturizer: binding with water, capable of holding up to 1,000 times its own weight in water
    - helps with collagen synthesis
    - aging = decrease in hyaluronic production of the body

MOISTURIZING

- Occlusives (petrolatum, lanolin & silicones, etc.): leave a film on the surface of the skin and seal in moisture

MOISTURIZING

- Intercellular substances (e.g., ceramides, cholesterol & fatty acids): work to hold skin cells together and maintain skin's youthful plumpness
  - The 2:4:2 ratio of natural lipids (ceramides, cholesterol & fatty acids, respectively)

MOISTURIZING ADVANCED

- cell-communicating or cell-signalling ingredients (*e.g.*, peptides, niacinamide, retinol & other antioxidants)
  - Cell's network of receptor sites for different substances → receive messages from a particular ingredient → pass along messages to nearby cells
  - cell-communicating ingredients tell a skin cell to behave better & stop other substances from telling the cell to behave badly.

RETINOIDS

- enhancing the skin's renewal process
- supporting the skin’s essential structural components responsible for skin firmness

→ acne treatment (unclog pore) & skin rejuvenation (boost cell turnover, stimulate collagen production, improve wrinkles and discoloration)

- The caveats: generally not recommended for younger people because their skin is less mature and more prone to irritation from potent active ingredients

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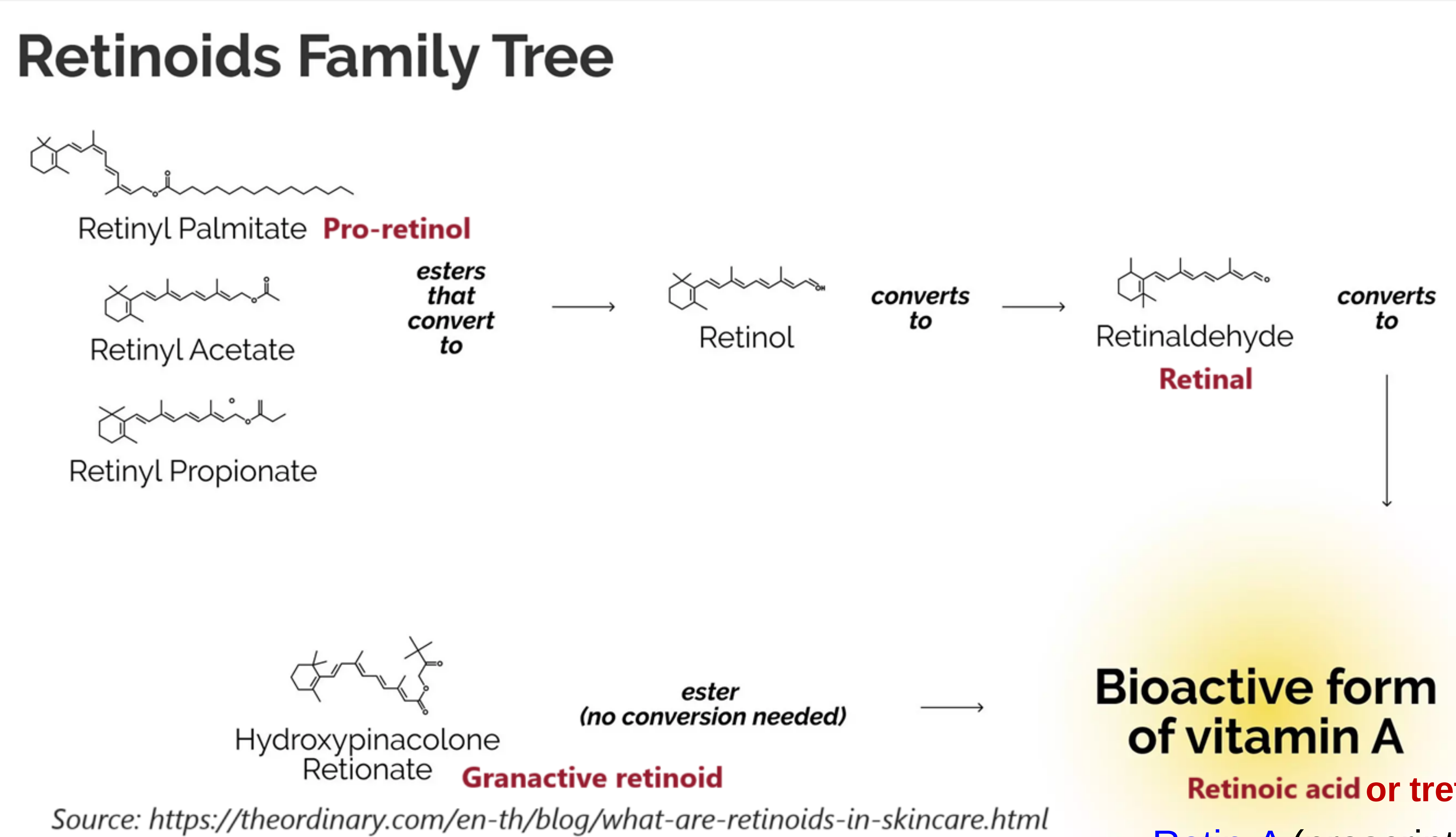
RETINOID FAMILY



or tretinoin

Retin-A (prescription)

Adapalene (OTC)

FOOD FOR THOUGHT

RETINAL - IS IT BETTER THAN RETINOL?

Sources: *https://chemistconfessions.com/blogs/retinal-is-it-better-than-retinol*

Retinaldehyde or retinal (with an A!) has become the latest, buzz-worthy retinoid on the market. In the process of retinoic acid synthesis, retinal would be the more efficient (aka more potent) cosmetic retinoid compared to retinol in getting those anti-aging benefits. You might even hear retinal claims such as "*10x more effective than retinol*" or even "*tretinoin strength without the tretinoin irritation.*" Bold claims!

While retinal isn't the newest kid on the block, its entry into skincare feels like it's only just beginning. In this retinal guide, we'll go over what retinal is, do these bold claims hold up, and who should be using retinal in their routine.

What's the difference between retinal vs. retinol?

Think of retinal and retinol as closely related molecular cousins. **Both of the molecules are in the retinoid family[1] and are key molecules in the retinoic acid synthesis pathway**.

Retinal is one step closer than retinol to converting to the retinoic acid form that your skin's retinoic acid receptors can interact with to kick off the anti-aging processes. On the other hand, retinol has to go through a 2 step-conversion. It's because of this shortened process, that retinal in theory, should be much more powerful than retinol. However, like all things skin science, there are caveats.

FOOD FOR THOUGHT

https://www.provenskincare.com/blog/granactive-retinoid-vs-retinol/

Granactive retinoid has been found to be effective in reducing the signs of aging with fewer side effects than traditional retinoids such as retinol [8]. It takes less time for your skin to break down the molecules to produce quicker results. In summation, it can be more tolerable for sensitive skin types [9] or acne-prone skin.

Other key differences to consider include:

- **Skin irritation:** Granactive retinoid has a low potential level for skin irritation and can be combined with other active ingredients like AHA [10]. However, retinol should be used on its own as other ingredients can counteract with the molecule to further exacerbate your skin condition.
- **Potency:** Retinol may take longer to see results. Granactive retinoid can provide the same, if not better, results, and is gentle on the skin [11].
- **Shelf life:** Granactive Retinoid has a longer shelf life, meaning it is more stable and less likely to break down when exposed to light or air. According to the Journal of Cosmetic Dermatology, retinol's potency can decrease overtime when stored in warm temperatures [12].

COMBINED USAGE

CAN YOU USE HYALURONIC ACID WITH RETINOL? *by* **Caroline C. Boyle**

In recent years, many have hailed the benefits of incorporating hyaluronic acid and retinol into your skincare routine, and the buzz around these ingredients isn't unfounded.

Studies have proven that hyaluronic acid creams support the production of collagen and improve skin elasticity. Retinol products have also exhibited positive results in studies — consistent use can help smooth wrinkles and promote an even skin texture. While the individual benefits of these ingredients are undeniable, can hyaluronic acid and retinol be used together?

What is hyaluronic acid?

Hyaluronic acid, also known as HA, is a **naturally-occurring substance,** a long-chain carbohydrate present in our body and our skin. It's commonly found as an active ingredient in skin and hair products, and it's also frequently used in the form of an injectable dermal filler that sculpts and restores volume to the face. It's a **powerful humectant**[1], and **while "acid" may be in the name, hyaluronic acid's "*main job is not to do with exfoliation, but actually it's to do with moisture,"*** says Dr. Aamna Adel, a dermatologist. When hyaluronic acid is applied to the face, it **attracts water molecules to the skin, locking in moisture and boosting hydration**.[2] Think of it like a sponge; as it pulls in moisture and absorbs it, it puffs up, helping to both hydrate and plump the skin.

COMBINED USAGE

- AHA/BHA: yes, probably not at the same time
  - Retinol stimulates cellular turnover from the deeper layers up *vs.* AHA/BHA shed dead skin cells in the uppermost layers
- Vitamin C: yes
  - fight free radicals → help protect retinol from oxidization
- Benzoyl peroxide: No
  - oxidize retinoids → less effective

Slide No **97**

Moisturizers are most effective when skin is still damp → apply shortly after cleansing

Retinol on dry ~~damp~~ skin ~~right before lights-out~~

layered lightest-to-thickest application, e.g., cleanser → serum → moisturizer → sunscreen

Slide No **98**

PURGING *vs.* BREAK-OUT

- acne breakouts due to cell turnover acceleration from certain skincare products → bring preexisting microcomedones to the surface sooner
- post-application of an ingredient (e.g., AHA or retinol) that speeds up your skin's natural cycle (the 28-day dead-cell-shed new-cell-born process) → a good sign that the product is working
- normal purging should NOT last more than six weeks

RETINOL ALTERNATIVE

BAKUCHIOL SKINCARE BENEFITS: EVERYTHING YOU NEED TO KNOW ABOUT THE NATURAL ALTERNATIVE TO RETINOL *by* **Amber Voller & Lucy Partington**

Billed as the 'gold standard' in skincare, retinol is proven to speed up our skin cells' turnover. Fine lines are reduced, pigmentation is treated and blackheads and whiteheads are minimized. In their place, expect plumpness, radiance and a more even skin tone and texture. But, like most things we love, retinol has a downside. All that turbo-regeneration action can come with a side serving of dryness, flaking, redness and inflammation.

But what if we told you that there was a **kinder alternative to retinol**? An ingredient that promised the same transformative powers only without the risk of irritation? Well, according to scientists, there is, and it comes in the form of **a natural, plant-based ingredient called bakuchiol** (pronounced *buh-koo-chee-all*).

Derived from the seeds and leaves of Eastern Asia's 'babchi' plant (officially the Psoralea corylifolia plant), bakuchiol has been a mainstay in traditional Ayurvedic and Chinese skin-healing treatments for centuries. It's long been loved for its **antioxidant, anti-inflammatory and antibacterial properties**.

Bakuchiol creams, oils and serums come in all kinds of textures and colours but the ingredient itself, like the flowers of the babchi plant, is a pale purple shade.

As Anjali Mahto, consultant dermatologist at Self London explains, **Bakuchiol has been shown to activate a number of chemical pathways in skin cells that ultimately lead to improved collagen production, decreased collagen breakdown and reduction of melanin (pigment) synthesis**. The overall net effect seems to be an **improvement in fine lines, wrinkles and pigmentation** that are commonly associated with the natural ageing process of the skin.

PACKAGING

- The proclivity of many active ingredients (retinoids, peptides, antioxidants, hydroquinone) for breaking down after air and light exposure
- packaging tips: ~~jar~~ *vs.* tube **<** pump → fridge & vacuum sealer to limit exposure to light and air

Slide No **101**

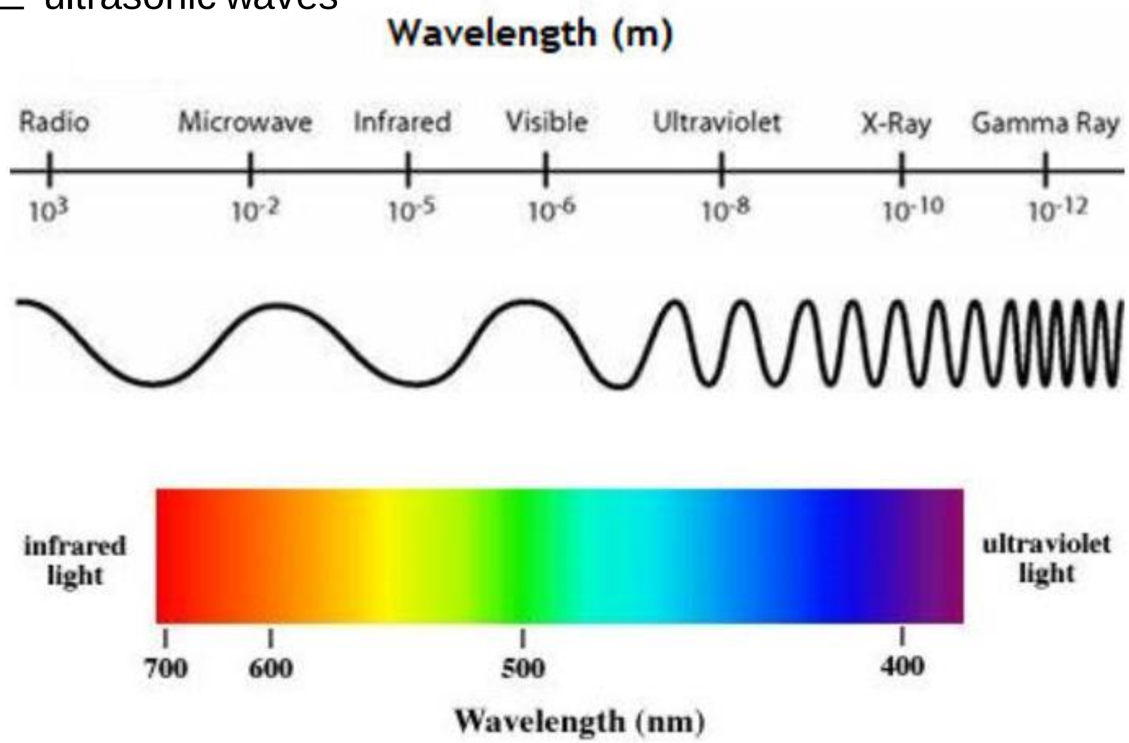
Type of Radiation	Frequency Range (Hz)	Wavelength Range
gamma-rays	10^{20} - 10^{24}	< 10^{-12} m
x-rays	10^{17} - 10^{20}	1 nm - 1 pm
ultraviolet	10^{15} - 10^{17}	400 nm - 1 nm
visible	4 - $7.5*10^{14}$	750 nm - 400 nm
near-infrared	$1*10^{14}$ - $4*10^{14}$	2.5 μm - 750 nm
infrared	10^{13} - 10^{14}	25 μm - 2.5 μm
microwaves	$3*10^{11}$ - 10^{13}	1 mm - 25 μm
radio waves	< $3*10^{11}$	> 1 mm
ultrasonic waves		

LLLT therapy

Laser re-surfacing

Thermage

HIFU



WAVE-BASED TECHNOLOGY

- Similar premises:
  - [**higher energy**] create damage to stimulate regeneration
  - [**lower energy**] energize the cells properly (in terms of both wavelength and intensity) to make them perform better
- Differences in
  - wave length (the "drug"), hence the penetration depth & targeted cells
  - intensity (the "dose"), hence the level of damage

WAVE-BASED TECHNOLOGY ISSUES

- Debatable correlations between wavelength and benefits?
- Debatable level of intensity: with increased energy, cells perform better or worse?
- The particularly controversial blue light: the ill-defined safe margin between acne treatment (short-term) *vs.* skin aging (long-term)
- Professional *vs.* Home versions
  - Relevant technologies confirmed by mainstream research
  - Devices approved by FDA?

Slide No 104

Are all microneedling products considered medical devices that are regulated by the FDA?

The FDA regulates microneedling products that are medical devices to make sure they are safe and work as claimed. Not all microneedling products are medical devices.

Microneedling products that are medical devices

The FDA has cleared microneedling devices for use as a treatment to improve the appearance of facial acne scars, facial wrinkles, and abdominal scars in patients aged 22 years or older.

Caveats: FDA-approved claims

Most of the cleared devices are pen-shaped, motorized and penetrate the skin in order to change the structure or function of the tissue below. Because these devices may reach nerves, blood vessels and other parts of the skin, the FDA recommends you go to a health care provider with special training in microneedling.

The FDA has not authorized any microneedling medical devices for over-the-counter sale.

Microneedling products that are NOT medical devices

Generally, if a microneedling product does not have longer needles or sharp needles that penetrate the skin and only claims to facilitate exfoliation of the skin or improve the appearance of skin, it would not be a medical device regulated by the FDA. A dermal roller with short, blunt needles that only claims to help remove dead skin and smooth and brighten your skin would be an example. These products are more commonly sold for use at home.

IF I MUST OWN A HIGH-TECH GADGET RIGHT NOW...

A moisture meter showing level of moisture and/or oiliness
The optimal range: 35%-60% → <35% should be alarming
→ check the effects of a new product on your skin

Slide No **106**

MY OWN ROUTINE

1. Cleanser: once or twice daily
2. Lip balm: daily
3. Physical exfoliant: as needed
4. Hyaluronic gel: daily
5. Retinol: every other day
6. Anti-acne gel: as needed
7. Sunscreen: as needed
8. Yogurt (occasionally)
9. (*considering*) antioxidant serum

POST-CLASS COMMUNICATION

- No advices on a particular product or skin condition
- No advices on problems that require doctor consultation
  - chronic skin conditions, e.g., eczema, psoriasis, & rosacea
  - complications of unsuitable skincare products

→ an educated user ≠ a dermatologist

FINAL NOTES

One-step-at-a-Time

Skincare products are not iPhones

STRUCTURES

- And I feel like that at this point, we talk about it ad nauseam [ON]
  - Sports commentators repeat the same phrases ad nauseam.
  - This topic has been discussed and analyzed ad nauseam.
- What exactly does it do in our bodies to wreak all that havoc.
  - Rioters caused havoc in the centre of the town.

STRUCTURES

- We evolved in a world that's saturated with UV light.
  - The shops were saturated with merchandise from the film.
- Digital devices emit only a fraction of that amount of radiation.
- [**literal**] radiate [N] *vs.* give off [I→N] *vs.* emit [F]
  - the heat radiated by the sun
  - The fire doesn't seem to be giving off much heat.
- [**figurative**] She radiates ~~emits~~ happiness and health.

STRUCTURES

- Topical antioxidants are "an absolute must".
- requirement [N] *vs.* must [I, **functional shift**] *vs.* requisite [F] *vs.* sine qua non [F]
  - Concentration was the first requirement for learning.
  - A university degree has become a requisite for entry into most professions.
  - Trust is a sine qua non for any counseling.
    - Reliability is a sine qua non for success.
  - Exercise is a must if you want to stay healthy.
    - a must-have phone | must-see movie | must-read book | must-do list [I→N]

STRUCTURES

- life expectancy | lifespan [N] *vs.* longevity [F]
  - They had longer life expectancies than their parents.
  - The natural lifespan of a pig is 10–12 years.
  - a diet that promotes longevity.
- This later year is not kind to [F] you.
  - Summer clothes are less kind to fuller figures.
  - Gentle cleansers are much kinder to your face skin than soaps.
- đời/số phận không bạc đãi ai?

STRUCTURES

- đời/số phận không bạc đãi ai?
  - fate/life has been kind to someone

STRUCTURES

- dilute solution with water
- dilute [N] *vs.* water down [I→N] *vs.* adulterate [N→F, ON] something with something else
  - The paint can be diluted with water to make a lighter shade.
  - No watered-down alcohol at this party.
  - The company was fined for adulterating its "100%" juices with water.

STRUCTURES

- "*One Mississippi, Two Mississippi...*" is a childish practice to approximate the passing of a second. It's traditionally used by children playing hide-and-seek to make sure the seeker counts long enough for others to hide.

  vs. The street lamps played hide-and-seek among the trees.

CLASS WRAP-UP

1. Tone: N *vs.* I *vs.* F
2. Register: S *vs.* W *vs.* A
3. Core message *vs.* ~~word-for-word~~
4. Cultural schemas & Linearity
5. Vocabulary recall *vs.* ~~vocabulary memorization~~
6. Linguistic tools for tone/register/nuance identification & differentiation
7. Content-based language learning

Nguyễn Thuỳ Dương
Từ buổi đầu đến h đây là chủ đề em thấy enjoy nhất, e ko còn quan tâm đến vấn đề học tiếng mà chỉ quan tâm đến kiến thức về da:)), hóng buổi sau quá ạ,

FOOD FOR THOUGHT

Thưa thầy,
Hôm nay trên lớp đại học của em, em được thầy giáo dạy dịch yêu cầu dịch bài viết này trên trang ĐSQ mỹ (http://vietnamembassy-usa.org/vietnam/foreign-policy). Sau đó khi giảng thì thầy giáo bảo những bài kiểu này phải dịch kiểu "để phân từ 2 lên đầu rồi dịch" và dịch "sao cho dễ hiểu nhất". Vấn đề là, cái bài này nó "Việt" tùm lum tung tóe. Ko theme, ko rheme, ko linear, no nothing, mà chắc chắn là thuần văn việt dịch sang anh. Dù vậy, thầy em vẫn dạy dịch như thường mà ko thay đổi hay thêm bớt gì, nói rằng dịch như vậy, dịch "để phân từ 2 lên rồi dịch", mới là "dịch (các bài policy) cho dễ hiểu nhất" và "truyền tải được chính xác ý người viết" ạ. Việc tương tự cũng xảy ra với cách thầy em dạy dịch việt-anh ạ.
Từ khi học TN của thầy, em không biết làm sao để sử dụng kiến thức TN trong btap đh nữa thầy ạ, khi thầy giáo em (thạc sĩ ngôn ngữ anh hẳn hoi) sửa những bài em dịch là thừa từ, dịch lung tung, ko sát ý tác giả :((
Thực sự em đang rối quá, làm theo kiểu thầy Vũ thì đúng, nhưng điểm kém, làm theo kiểu thầy giáo, thì chả khác gì google translate, nhưng lại có điểm :((((
Can you lend me some word of advice?

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